

Project

1. Team size: 3-5 members.
2. Three presentations: proposal, interim and final.
3. Optional topics: (Robotic arm and dexterous hand are encouraged, and can be combined)
 - (1) Robotic arm
 - (2) Dexterous hand
 - (3) Facial robot
 - (4) Biped robot

(Choose one topic, design and fabricate the robot by yourselves, and then try some applications.)

Please determine team members in the third week.

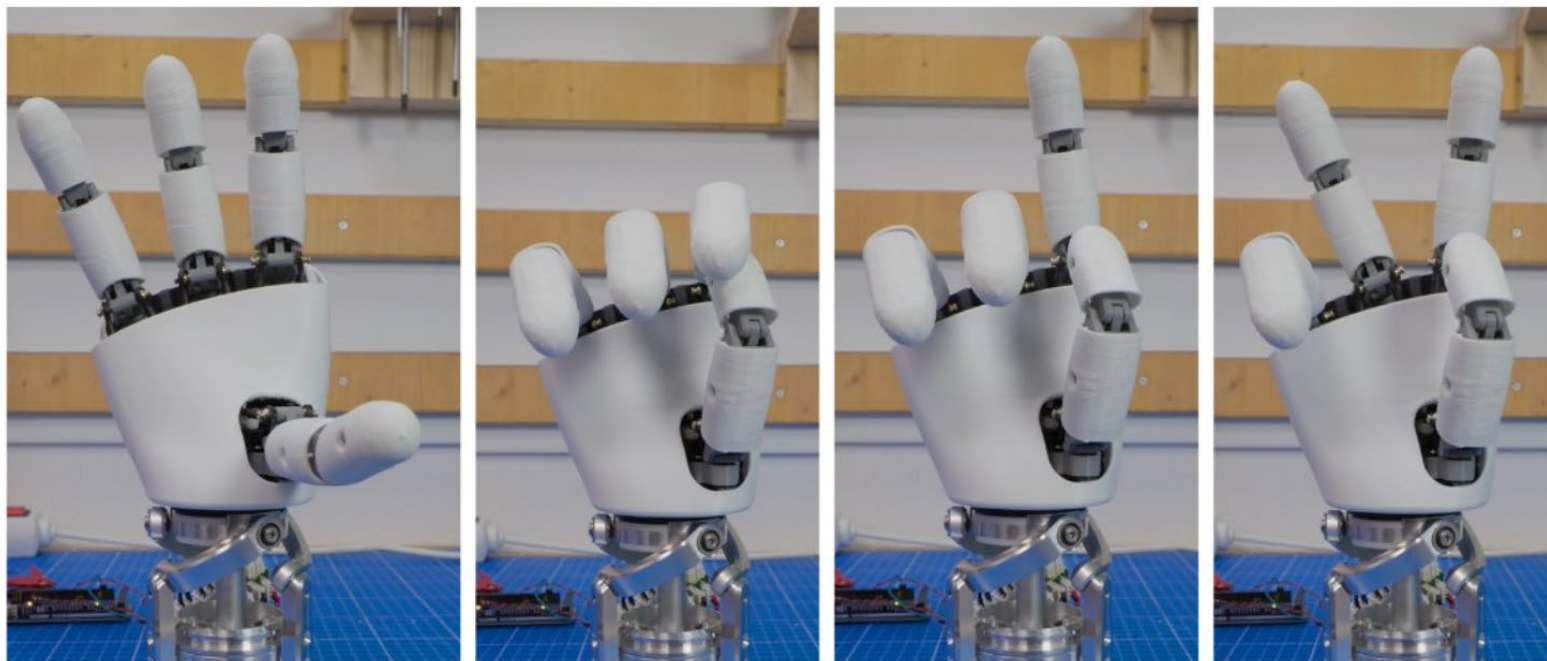
Schedule: (The initial plan could be adjusted, there will be detailed announcements before each pre.)

Proposal presentation	Feb 3, 6:00-9:00pm
Interim presentation	April 4, 6:00-9:00pm
Final presentation	May 14, 6:00-9:00pm
Final report submission on BB	May 16, 23:59pm

Dexterous hands

1.

<https://github.com/pollen-robotics/AmazingHand>

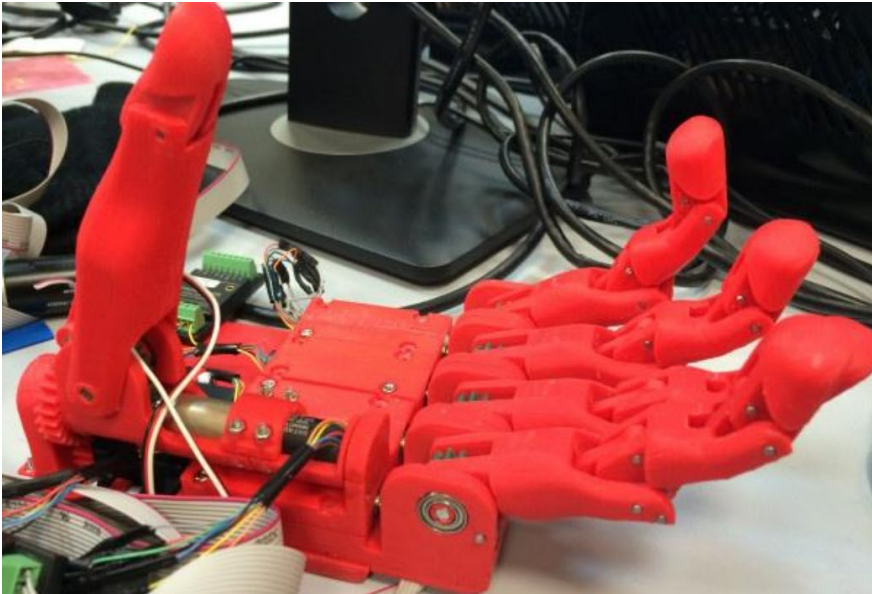


- 8 dofs humanoid hand with 4 fingers
- 2 phalanxes per finger articulated together
- flexible shells almost everywhere
- All actuators inside the hand, without any cables
- 3D printable
- 400g weight
- low-cost (<200€)
- open-source

2.

https://github.com/alarisnu/alaris_hand

An Open-Source Mechanical Design of ALARIS Hand: A 6-DOF Anthropomorphic Robotic Hand, 2021
IEEE International Conference on Robotics and Automation (IEEE ICRA 2021), China, May 2021.



3.

<https://ruka-hand.github.io>

RUKA: Rethinking the Design of Humanoid Hands with Learning, (RSS2025)



RUKA outperforms other hands in strength tests.

A Payload



B Slip Test



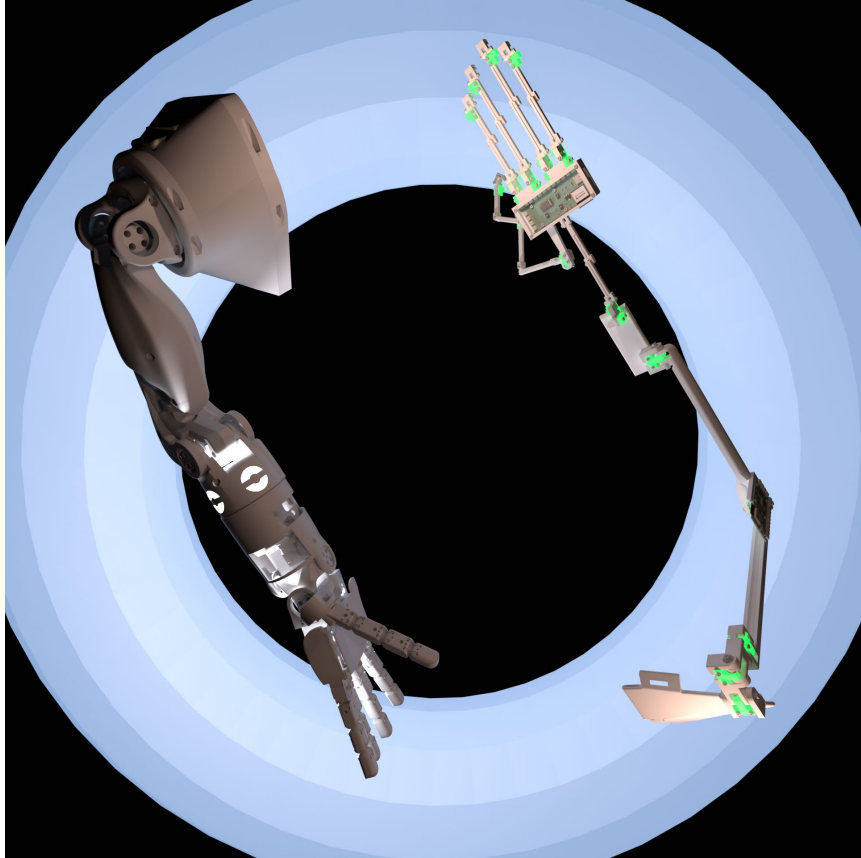
C Pinch Test



Objects:

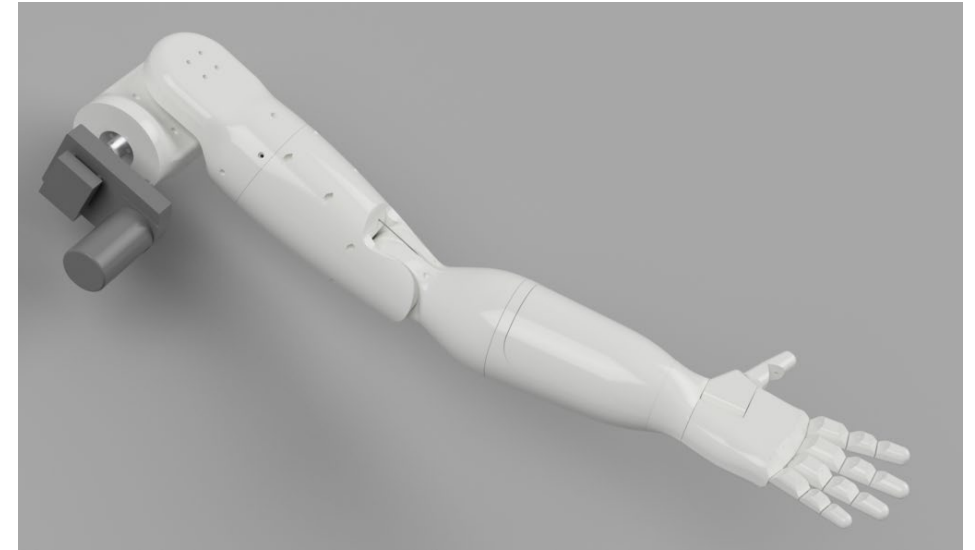
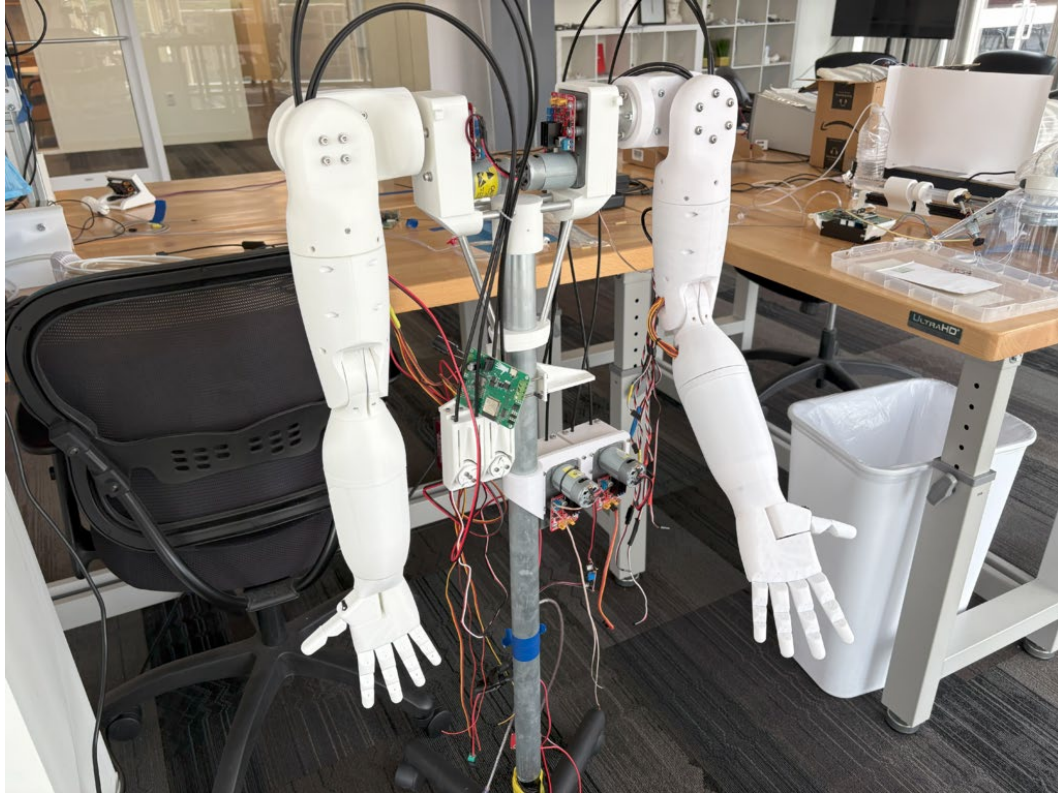
1. Design and fabricate a dexterous hand capable of imitating hand gestures and performing basic grasping functions.
2. After fabrication, the hand can be mounted on another group's robotic arm to enable grasping functionality. Moreover, it can be integrated with vision systems to achieve human-hand-pose imitation and any other related capabilities you want to realize.

Robot arm



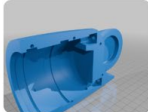
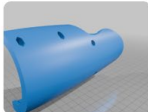
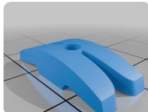
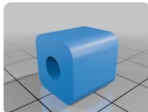
..	
3dprint files	add scs spool
Artwork	add separate Humanoid and Arm sections
PCB	add separate Humanoid and Arm sections
STEP	separate glove from exoskeleton
code	update exo code
.DS_Store	add teleop code
BOM.md	Modify spring specifications in BOM.md
ChangeLog	add separate Humanoid and Arm sections
README.md	add separate Humanoid and Arm sections

<https://github.com/TheRobotStudio/HOPEJr/tree/main>



<https://youtu.be/r62dPoS-24s>

<https://www.thingiverse.com/thing:7111274/files>

	BicepA.stl Size: 7 mb Updated: 08-06-2025 Downloads: 110 Open in Cura ... ↓
	BicepB.stl Size: 9 mb Updated: 08-06-2025 Downloads: 103 Open in Cura ... ↓
	Butterfly.stl Size: 770 kb Updated: 08-06-2025 Downloads: 95 Open in Cura ... ↓
	CableSeat.stl Size: 580 kb Updated: 08-06-2025 Downloads: 88 Open in Cura ... ↓

OpenArm



Repository	Documentation	License	Description
openarm	General Docs	Apache-2.0	Main project repository with ideas, issues, and feature requests
openarm_hardware	Hardware Docs	CERN-OHL-S-2.0	Complete CAD data: STL files, STEP files, Fusion 360 assemblies
openarm_description	Description Docs	Apache-2.0	Robot description files with URDF/xacro for simulation
openarm_can	CAN Docs	Apache-2.0	CAN control library for low-level motor communication
openarm_ros2	ROS2 Docs	Apache-2.0	ROS2 integration packages and nodes
openarm_teleop	Teleop Docs	Apache-2.0	Teleoperation packages with unilateral and bilateral control
openarm_isaac_lab	Isaac Docs	Apache-2.0	Isaac Lab simulation environment and training tasks

<https://github.com/enactic/openarm>

Core Project Objectives

1. **Basic Objectives:** Teams independently complete the construction of a 6-degree-of-freedom (recommended) 3D-printed robotic arm (including servo drive and mechanical structure assembly), realize kinematic modeling (DH parameter modeling), forward/inverse kinematics algorithm programming, and physical verification (e.g., fixed-point grasping, trajectory planning demonstration).
2. **Extended Objectives:** Integrate cutting-edge technologies (VLA, visual recognition, AI planning, etc.) to endow the robotic arm with intelligent functions, fostering interdisciplinary integration capabilities and innovative thinking.